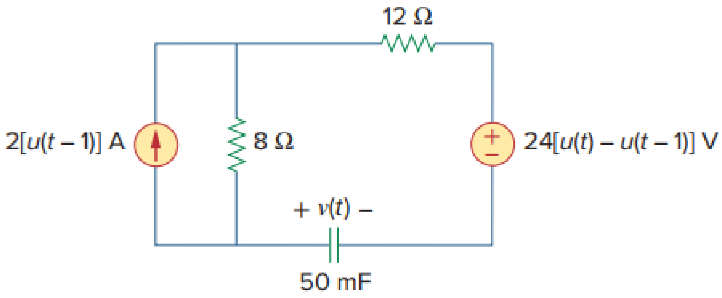
Chapter 7, Problem 47P

Problem

Determine $v(t)$ for $t > 0$ in the circuit of Fig. 7.114 if $v(0) = 0$.

Fig. 7.114:



Step-by-step solution

Step 1 of 8

Refer to circuit diagram in Figure 7.114 in the textbook.

It is known that,

$$u(t)=\begin{cases}0 & t<0\\ 1 & t>0\end{cases}$$

$$u(t-1)=\begin{cases}0 & t<1\\ 1 & t>1\end{cases}$$

$$u(t)-u(t-1)=\begin{cases}0 & t<0\\ 1 & 0<t<1\\ 0 & t>1\end{cases}$$

Step 2 of 8

For $t > 0$, the value of current source is zero and value of voltage source is 24 V. The circuit diagram for $t > 0$ is shown in Figure 1.

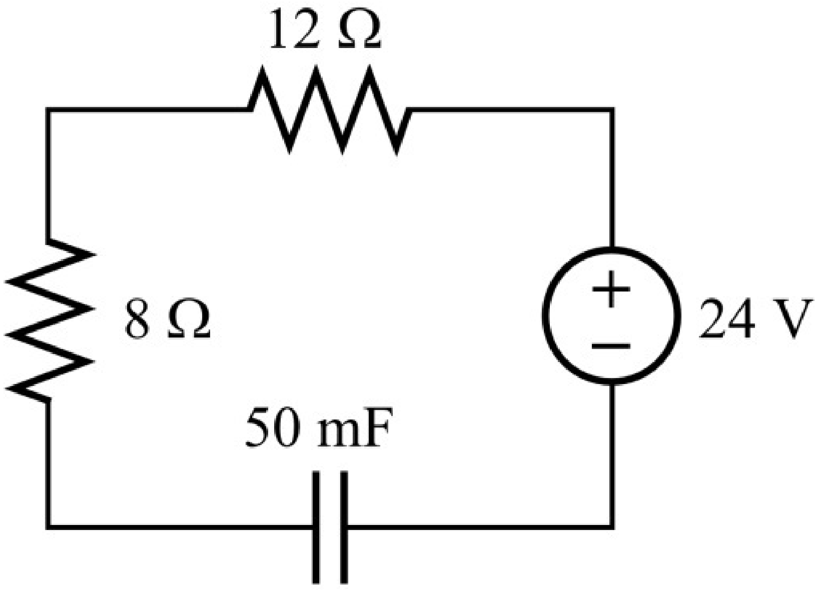


Figure 1

Step 3 of 8

Consider that the circuit in that position for a long time the circuit reached steady state and capacitor acts like an open circuit. The circuit diagram at $t = \infty$ is shown in Figure 2.

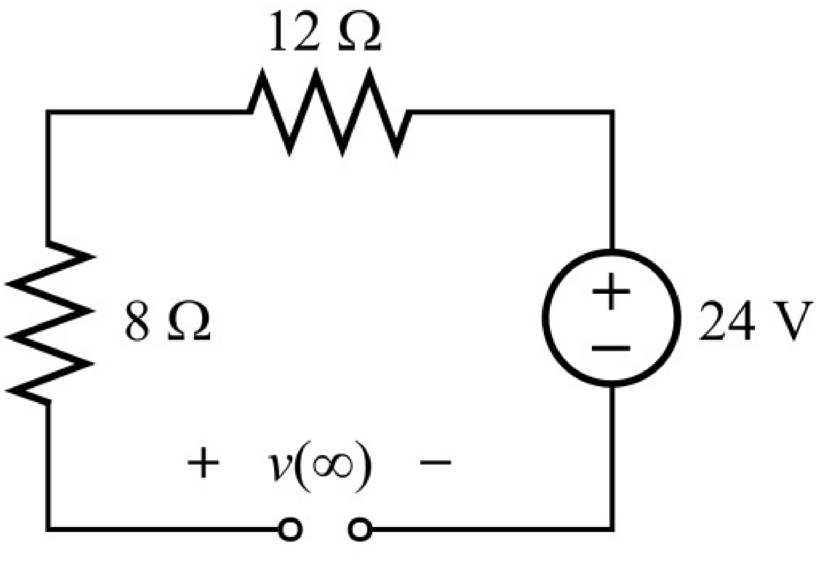


Figure 2

Step 4 of 8

Calculate the voltage across the capacitor at $t = \infty$.

$$v(\infty) = 24 \text{ V}$$

The initial voltage across the capacitor is,

$$v(0) = 0 \text{ V}$$

Calculate the equivalent resistance.

$$\begin{aligned} R_{eq} &= 8\ \Omega + 12\ \Omega \\ &= 20\ \Omega \end{aligned}$$

Calculate the time constant.

$$\begin{aligned} \tau &= R_{eq}C \\ &= (20)(50\times10^{-3}) \\ &= 1 \text{ sec} \end{aligned}$$

Step 5 of 8

Calculate the voltage across the capacitor for $t > 0$.

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$$

Substitute 0 V for $v(0)$, 24 V for $v(\infty)$ and 1 sec for τ.

$$\begin{aligned} v(t) &= 24 + [0 - 24]e^{-\frac{t}{1}} \\ &= 24(1 - e^{-t}) \text{ V} \end{aligned}$$

Calculate the voltage across the capacitor at $t = 1 \text{ s}$.

$$\begin{aligned} v(1) &= 24(1 - e^{-1}) \text{ V} \\ &= 24(1 - 0.3679) \text{ V} \\ &= 24(0.632) \text{ V} \\ &= 15.17 \text{ V} \end{aligned}$$

Step 6 of 8

At $t = 1 \text{ s}$, 2 A source comes to active and 24 V source becomes inactive. The circuit for $t > 1 \text{ sec}$ is as shown in Figure 3.

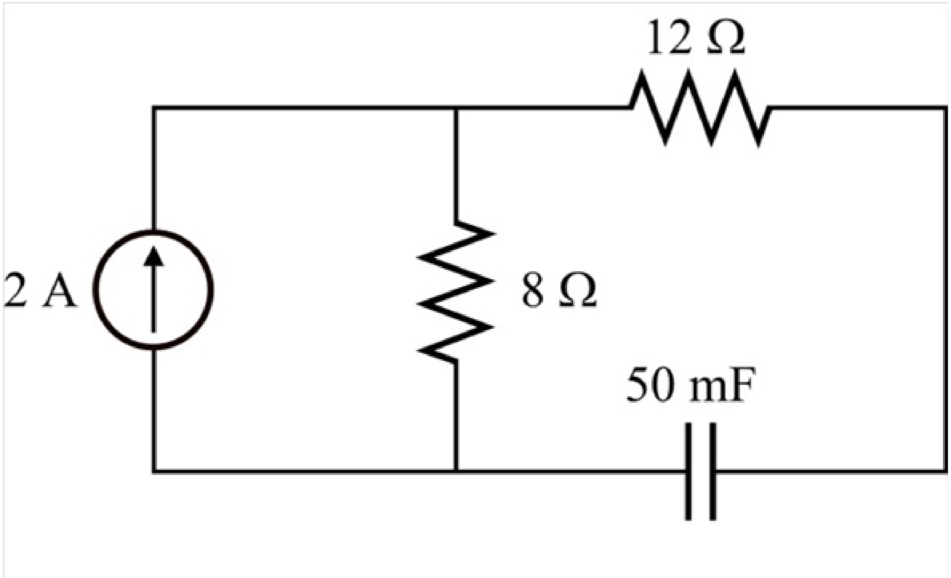


Figure 3

Consider that the circuit is in that position for a long time the circuit and reached steady state and capacitor acts like an open circuit. The circuit diagram at $t = \infty$ is shown in Figure 4.

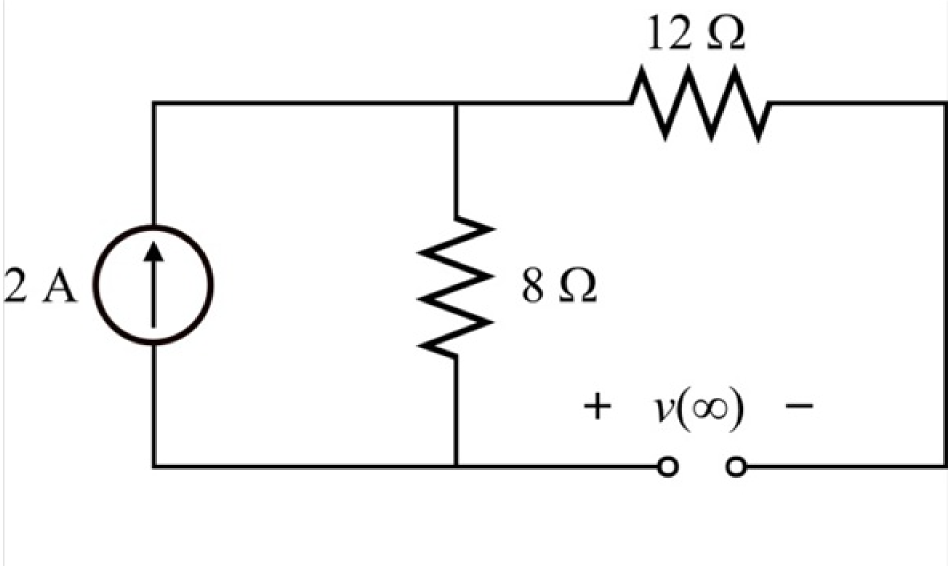


Figure 4

Calculate the voltage across the capacitor at $t = \infty$.

$$v(\infty) = (2\text{ A})(8\ \Omega)$$
$$= 16\text{ V}$$

Calculate the voltage across the capacitor for $t > 1$ is,

$$v(t) = v(\infty) + \left[v(1) - v(\infty)\right] e^{-\frac{t-1}{\tau}}$$

Substitute 15.17 V for $v(1)$, 16 V for $v(\infty)$ and 1 sec for τ.

$$v(t) = 16 + \left[15.17 - 16\right] e^{-\frac{t-1}{1}}$$
$$= 16 - 0.83e^{-(t-1)}\text{ V}$$

Therefore, the voltage across the capacitor is,

$$\boxed{v(t) = \begin{cases} 24\left(1 - e^{-t}\right)\text{ V} & 0 < t < 1\text{ s} \\ 16 - 0.83e^{-(t-1)}\text{ V} & t > 1\text{ s} \end{cases}}.$$